

TECHNOLOGICAL UNIVERSITY DUBLIN
City Campus - Grangegorman

TU856 - BSc. Degree in Computer Science

Year 3

SEMESTER 1 EXAMINATIONS 2025/2026

CMPU3051 Introduction to Artificial Intelligence

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Exam Duration: 2 hours

Instructions To Candidates:

Attempt 4 out of 5 questions. All questions carry equal marks.

Special Instructions /Handouts/ Materials Required: None

1 (a) Explain what is meant by *state space search*.

In state space search, discuss the difference in performance between depth first and breadth first search. Assume each state has **b** successor or branching states. Is there a way to combine the advantages of both?

(9 marks)

(b) Represent the following statements in Predicate Calculus:

- Lucy is a professor
- All professors are people
- John is the dean
- Deans are professors
- All professors consider the dean a friend or don't know him
- Everyone is a friend of someone

(8 marks)

(c) Prolog uses backward chaining. Explain what this means with the help of a proof tree in terms of the following Prolog program when the Prolog interpreter is asked to prove **k (X)**. Make reference to *unification* in your answer.

f (a) .

f (b) .

g (a) .

g (b) .

h (b) .

k (X) :- f (X) , g (X) , h (X) .

(8 marks)

2 (a) Write down 2 versions of Bayes' rule.

Given the following:

1% of people have a certain genetic defect.

90% of tests for the gene detect the defect (true positives).

9.6% of the tests are false positives.

Use Bayes' rule to calculate the odds that person actually has the genetic defect if they get a positive test result.

(9 marks)

- (b)** Give three reasons why Bayesian reasoning is difficult to use in an expert system. Also, briefly describe two advantages in using certainty factors as opposed to Bayesian reasoning.

Briefly mention two well-known historical expert systems, one of which used Bayesian reasoning and the other which used certainty factors.

(8 marks)

- (c)** Given the forecasting rules below, show how they would fire to forecast tomorrow's weather when provided with the following information: there is rain today and the rainfall is low with a certainty factor of 0.8, and also it is cold with a certainty factor of 0.9. This formula may be useful:

$$cf(cf_1, cf_2) = cf_1 + cf_2 \times (1 - cf_1)$$

Rule: 1

if today is rain

then tomorrow is rain {cf 0.5}

Rule: 2

if today is dry

then tomorrow is dry {cf 0.5}

Rule: 3

if today is rain

and rainfall is low

then tomorrow is dry {cf 0.6}

Rule: 4

if today is rain

and rainfall is low

and temperature is cold

then tomorrow is dry {cf 0.7}

Rule: 5

if today is dry

and temperature is warm

then tomorrow is rain {cf 0.65}

Rule: 6

if today is dry

and temperature is warm

and sky is overcast

then tomorrow is rain {cf 0.55}

(8 marks)

- 3 (a) Express the following function and Predicate Calculus statements in Prolog:

$\text{minincome}(Y) = 15000 + 4000 * Y$

$\text{savings_account}(\text{adequate}) \wedge \text{income}(\text{adequate}) \Rightarrow \text{investment}(\text{stocks}).$

$\forall X \text{ earnings}(X, \text{steady}) \wedge \forall Y \text{ dependents}(Y) \wedge \text{greater}(X, \text{minincome}(Y))$
 $\Rightarrow \text{income}(\text{adequate}).$

(5 marks)

- (b) Write a recursive Prolog predicate **canGoFromTo(A, B)** to express the following:
One can travel from A to B if there is a bus or train from A to B. Or one can travel from A to B if there is a bus or train from A to C and one can travel from C to B.
(5 marks)
- (c) Write a Python function `myDotProd(a,b)` which computes the dot-product of 2 vectors of equal size using a loop. You are not to use the `numpy.dot()` operation.
(5 marks)
- (d) Write Python code for a simple perceptron decision function **buyCocktail(x)**, which decides on whether to buy an expensive cocktail or not based on the following inputs where $x = [x_1, x_2, x_3]$.

$x_1 = 1$ means you have plenty of cash, $x_1 = 0$ means you do not. $x_2 = 1$ means you want to impress a new girlfriend, 0 means you don't. $x_3 = 1$ means it's a busy exciting night on the town, 0 means it is not.

Choose the weights and threshold (or bias) so that if you want to impress someone, you will buy cocktails no matter what. However, if you are not out to impress but it's an exciting night and you have plenty of cash, you will buy a cocktail. Otherwise you will not buy a cocktail.

(5 marks)

- (e) Suppose you were to train a perceptron to behave as described in part(d) instead of handcrafting the weights. Write down an appropriate list of input training vectors and the corresponding list of desired outputs which could be used to train such a perceptron.
(5 marks)

- 4 (a) What is a perceptron? Draw a diagram for a single two-input perceptron. Explain what is meant by linear separability. How many categories can it classify inputs into? Mention any issues involved in using a perceptron to compute logical AND, OR and XOR. (7 marks)
- (b) Illustrate the single perceptron training algorithm for an OR gate by computing the weight adjustments for the first two epochs in Table 4(b) below. (5 marks)
- (c) Distinguish between a sigmoid neuron and a perceptron, and outline the difference in the training formula for Δw_i .

Why might it be preferable to use sigmoid neurons in a neural network rather than perceptrons? (8 marks)
- (d) Provide a Python code segment which implements the perceptron training algorithm. (5 marks)

Perceptron Training for logical OR

Threshold $\theta = 0.2$, learning rate $\alpha = 0.2$.

Epoch	Inputs		Desired output Y_d	Weights		Weighted Sum z	Actual Output a	Error e	Weight adjustments	
	x_1	x_2		w_1	w_2				Δw_1	Δw_2
1	0	0	0	0.4	-0.3					
	0	1	1							
	1	0	1							
	1	1	1							
2	0	0	0							
	0	1	1							
	1	0	1							
	1	1	1							
3	0	0	0							

Table 4(b)

5 (a) Suggest a network architecture appropriate for the MNIST dataset and explain why the cross-entropy cost function might be preferable to the mean-squared-error when the sigmoid activation function is used. (5 marks)

(b) Explain the problem of overtraining a neural network and how it could be alleviated. How would you partition the MNIST dataset to address this problem? (5 marks)

(c) Briefly explain what gradient descent means and how it relates to the cost or loss function. Explain what each of the following symbols represents and the relationship between them in gradient descent.

$$\Delta w_{ij}^l \text{ and } \frac{\partial C}{\partial w_{ij}^l} ?$$

(5 marks)

(d) At the core of the backpropagation algorithm are the matrix formulae:

$$\frac{\partial C}{\partial w_{jk}^l} = \delta_j^l a_k^{l-1}$$

$$\delta^L = (a^L - y) \odot \sigma'(z^L) \quad \delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

Briefly explain what each term means and which one in particular relates to backpropagation.

Also, for a three layer network, write a Python code fragment which includes each of these formulae.

(10 marks)